

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 2 -Technical Presentation

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 2 Technical Presentation
Weightage	30% of CIA	Total Marks	100
CO1 Statement	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
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Section/Batch	Slot C /2025	Date	06-10-2026

Code of Conduct:

I, CH.V.N.V.S SRIRAM , certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
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Technical Presentation
CSA0314 - DATA STRUCTURES

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Problem & Requirement

- Problem:
- A messaging application temporarily stores recent chat messages for quick access.
- Messages are added as new chats arrive and older messages may need to be removed.
- The application should access the most recent messages quickly.
- Requirements:
- Store recent messages temporarily.
- Add new messages efficiently.
- Remove the oldest messages when storage is full.
- Access recent messages quickly.

Suitable Data Structure: Circular Queue

- A Circular Queue is a linear data structure that follows FIFO (First In, First Out).
- Front → points to the oldest message.
- Rear → points to the newest message.
- When the queue becomes full, the oldest message can be removed as a new message is added.
- The circular arrangement reuses available positions efficiently.

Operations & Efficiency

- Enqueue (add message): $O(1)$ time.
- Dequeue (remove oldest message): $O(1)$ time.
- Peek/access the front message: $O(1)$ time.
- Checking the newest message at the rear: $O(1)$ time.
- Space Complexity: $O(n)$, where n is the maximum number of recent messages stored.
- Therefore, a circular queue provides fast operations with fixed temporary storage.

Why Circular Queue is Appropriate?

Data Structure	Suitability	Reason
Array	Moderately suitable	Fast access, but managing oldest messages needs extra logic.
Linked List	Suitable	Easy insertion/deletion, but extra memory for links.
Stack	Less suitable	LIFO does not match oldest-message removal.
Queue	Highly suitable	FIFO matches recent-message storage and removal.
Circular Queue	Highly suitable	FIFO + efficient reuse of fixed storage.

Advantages of Circular Queue

- Maintains messages in chronological order.
- New messages are added at the rear.
- Oldest messages are removed from the front.
- Insertion and deletion can be performed in $O(1)$ time.
- Fixed-size storage prevents unlimited memory growth.
- Circular reuse avoids wasting array positions.
- Suitable for temporary caches, recent-message lists, and streaming data.

Example

- Suppose the application stores only 5 recent messages.
- $M1 \rightarrow M2 \rightarrow M3 \rightarrow M4 \rightarrow M5$
- When M6 arrives, M1 (the oldest) is removed.
- Queue becomes: $M2 \rightarrow M3 \rightarrow M4 \rightarrow M5 \rightarrow M6$
- When M7 arrives, M2 is removed.
- Queue becomes: $M3 \rightarrow M4 \rightarrow M5 \rightarrow M6 \rightarrow M7$
- This keeps only the latest messages while using fixed storage.

Conclusion

- A Circular Queue is appropriate for temporarily storing recent chat messages.
- Messages are maintained in FIFO order.
- New messages are inserted at the rear and the oldest messages are removed from the front.
- Enqueue, dequeue, and front/rear operations can be $O(1)$.
- Space complexity is $O(n)$ for n stored messages.
- It provides simple and efficient management of a fixed-size recent-message cache.

THANK YOU